

Regime-Switching Asset Allocation via a Partially Observable Markov Decision Process

Type: New Application

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Abstract

We model dynamic stock-bond allocation as a Partially Observable Markov Decision Process (POMDP) whose latent state is the prevailing macro-financial regime. A two-state Gaussian Hidden Markov Model (HMM) is fit by Baum–Welch on the VIX and the 10Y–3M Treasury term spread; a recursive Bayesian filter maintains a belief $\mathbf{b}_t$ over the latent regime, updated each month from the new observation. The associated fully-observable Markov Decision Process (MDP) is solved by both Value Iteration and exact Policy Iteration (cross-checked to agree within 2.6×10^{-6}). The QMDP approximation [12], which we derive explicitly from the belief-space Bellman equation as a one-step value-function approximation, selects the weight that maximises the belief-weighted action value $\sum_s \mathbf{b}_t(s) Q^*(s, \mathbf{a})$ at each rebalance. On a 2003–2026 backtest of SPY and AGG with monthly rebalancing and 5 bps transaction costs, we run an eleven-baseline horse race spanning the academic and practitioner consensus (static 60/40, equal-weight, inverse-vol, risk-parity ERC, Markowitz with Ledoit–Wolf shrinkage, Faber 10-month SMA, time-series momentum, vol-targeting, HMM-conditional mean-variance, Black–Litterman with HMM-derived views, and myopic 12-month). At the canonical CRRA $\gamma=2$, the underlying MDP is genuinely regime-differentiated: the optimal policy is 100% equity in the bull regime and 100% bonds in the bear regime, and the resulting QMDP policy attains a Sharpe ratio of **1.18** (4th of twelve), comfortably beating the static 60/40 (0.81) and essentially tying the HMM-conditional mean-variance baseline (1.20), while cutting maximum drawdown from -34% to -14% . The belief filter is genuinely recursive: $P(\text{bear} \mid \mathbf{o}_{1:t})$ tracks every NBER recession and the 2015–16 and 2022 stress episodes, and a single VIX spike flips the belief from 0.3% to 98.7% bear in one filter step. The honest verdict is that a correctly-specified POMDP regime overlay is a competitive risk-adjusted allocator that beats the static benchmark, though it does not overtake the practitioner-consensus trend rule (Faber, 1.68). Regime differentiation is robust across CRRA levels, observation cohorts, and three economically distinct sub-periods; expanding-window walk-forward refit [17] further improves it. We position QMDP within Powell’s [19] taxonomy as a value-function approximation with one-step direct lookahead.

1 Introduction

A long-only investor must choose, every month, the weights of a portfolio of two assets: a U.S. equity index (SPY) and a U.S. aggregate-bond index (AGG). The textbook 60/40 split is a *static* allocation that implicitly assumes the joint return distribution is stable. Markets violate this in two specific ways: returns are conditionally heteroskedastic (calm decades broken by short crises), and

stress periods exhibit higher cross-asset correlations so bond diversification erodes when it is most needed. In 2022 a 60/40 portfolio lost roughly 17% as both legs sold off under persistent inflation, motivating tactical regime-aware overlays.

Research question. We test the implicit claim of every regime-switching paper: *Can a fully decision-theoretic regime overlay (formalised as a POMDP, fit to public macro-financial data, solved by a tractable belief-state heuristic) beat the static 60/40 benchmark out-of-sample after realistic transaction costs?* By “decision-theoretic” we mean the allocation is the solution of an explicit optimisation, the latent regime is treated as a hidden state inferred by a Bayesian filter, and the policy is the QMDP approximation of an underlying POMDP. Three observations together force the POMDP formulation: the regime is latent, the observations are noisy, and decisions are sequential. Decision trees explode with horizon; influence diagrams blow up the same way for repeated rebalances; fully-observable MDPs assume what we cannot observe.

Contributions. (i) an explicit derivation of the QMDP rule from the belief-space Bellman equation, together with a fully-specified recursive belief filter and a worked numerical example (Sections 3.1 and 3.3), the methodological core; (ii) an eleven-baseline horse race spanning the academic and practitioner consensus (Section 5.1), in which the QMDP overlay beats the static 60/40 and ties the HMM-conditional mean-variance baseline; (iii) robustness evidence that the regime-differentiated policy holds across CRRA levels (Section 5.2), observation cohorts (Section 5.3), and three economically distinct sub-periods (Section 5.5), with expanding-window walk-forward refit [17] a further improvement (Section 5.4); (iv) a Powell-taxonomy positioning of QMDP as a VFA + one-step DLA; (v) a reproducible public repository `takahoo/ENGS177_Final_Project` with 14 data series, 12 strategies, and 13 experiment scripts.

2 Background

Regime switching in finance. The Markov-switching model is due to Hamilton [7] who applied a two-state model to U.S. GDP. Ang and Bekaert [1] demonstrated that ignoring regimes produces welfare losses in international allocation, but their gain mechanism is concentrated in the cash-switching margin: “costs of ignoring regimes are small for all-equity portfolios.” Guidolin and Timmermann [6] formalised multi-asset allocation under a 4-state HMM and reported certainty-equivalent gains of 1.5–3.0% per annum. Nystrup et al. [17] address the look-ahead-bias problem via online walk-forward refit. Tu [23] examines whether regime switching matters once model, parameter, and regime uncertainty are jointly accounted for; the answer is qualified. A central methodological point of our study is that whether the regime label moves the optimal policy depends critically on the regime-conditional reward being estimated in the same space the HMM is calibrated in: with a correctly-applied belief filter, the label moves the policy even at the canonical low risk aversion ($\gamma=2$), a two-risky-asset universe notwithstanding.

POMDP framework. A Partially Observable MDP [10] assumes the agent observes only a noisy function of the latent state and maintains a probability distribution (belief) over the state. Exact POMDP solution is PSPACE-hard; the QMDP approximation [12] projects the belief onto the action-value function of the fully-observable MDP. We use QMDP because (i) it is exact when the belief is a Dirac measure, (ii) it reduces to a standard infinite-horizon MDP solver, and (iii) in our setting there is no action that improves observability of the regime (we observe VIX and the

term spread regardless of allocation), so the well-known weakness of QMDP (failure to motivate information-gathering actions [8]) does not bind. Point-based value iteration [18] is a tighter alternative but unnecessary on our 2-state model with a 1-dimensional belief simplex.

Powell taxonomy positioning. Powell [19] unifies sequential decision-making into four meta-classes: policy function approximations (PFAs), cost function approximations (CFAs), value function approximations (VFAs), and direct lookahead approximations (DLAs). QMDP fits naturally as a *VFA* in which the belief-state value function $V(\mathbf{b})$ is approximated by the linear lower envelope $V_{\text{QMDP}}(\mathbf{b}) = \max_a \sum_s b(s) Q_{\text{MDP}}^*(s, a)$, combined with a *one-step DLA* at decision time. This sentence is the project’s theoretical anchor: we are using the MDP value function as a VFA on the belief MDP and selecting actions via a one-step lookahead.

Practitioner consensus. The empirical winner of published two-asset tactical horse races (Faber [5], Hurst–Ooi–Pedersen “Century of Evidence” [9], Moskowitz–Ooi–Pedersen [16], Moreira–Muir [15]) is consistently *time-series momentum with volatility targeting and cash fallback*, reporting Sharpe 0.85–1.10 on SPY+AGG 1990–2024. This is the empirical bar our QMDP must clear.

3 Methods

3.1 POMDP formulation

Definition 1 (POMDP). A POMDP is the tuple $(\mathcal{T}, \mathcal{S}, \mathcal{A}, T, \mathcal{O}, O, R, \lambda)$ where $\mathcal{T} = \{0, 1, \dots\}$ is the set of monthly epochs; $\mathcal{S} = \{s_1, \dots, s_K\}$ is the finite set of latent regimes; $\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_M\}$ is the discrete grid of long-only stock-bond weights; $T(s' | s)$ is the HMM transition matrix (action-independent because regimes are exogenous to portfolio choice); $\mathcal{O} = \mathbb{R}^d$; $O(\mathbf{o} | s) = \mathcal{N}(\mathbf{o}; \boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ is the Gaussian emission; the single-period reward is the expected CRRA utility of next month’s *gross* portfolio return,

$$R(s, \mathbf{a}) = \mathbb{E}_{\mathbf{r}|s}[U_\gamma(1 + \mathbf{a}^\top \mathbf{r})], \quad U_\gamma(w) = \frac{w^{1-\gamma}-1}{1-\gamma},$$

with $\gamma=2$; and $\lambda = 0.95$. Transaction costs ($c=5$ bps of $\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_1$) are *not* embedded in R ; they are charged in the backtest wealth accounting (Section 3.5), because the previous weight $\mathbf{a}_{t-1}$ is not part of the regime state, and folding it in would require augmenting the state space, which we deliberately do not do.

Recursive belief filter. The agent never observes the regime; it maintains a belief $\mathbf{b}_t \in \Delta^{K-1}$ and updates it each month from the new observation $\mathbf{o}_t$ by one step of the standard forward (Bayes) filter, in two stages:

$$(\text{predict}) \quad \hat{\mathbf{b}}_t(s') = \sum_s T(s' | s) \mathbf{b}_{t-1}(s), \tag{1}$$

$$(\text{update}) \quad \mathbf{b}_t(s') = \frac{O(\mathbf{o}_t | s') \hat{\mathbf{b}}_t(s')}{\sum_{s''} O(\mathbf{o}_t | s'') \hat{\mathbf{b}}_t(s'')}. \tag{2}$$

The predict step advances last month’s belief through the regime dynamics (it nudges the belief toward the other regime by the small switch probability); the update step multiplies by the Gaussian observation likelihood $O(\mathbf{o}_t | s') = \mathcal{N}(\mathbf{o}_t; \boldsymbol{\mu}_{s'}, \boldsymbol{\Sigma}_{s'})$ and renormalises. We initialise $\mathbf{b}_0$ at the

stationary distribution of T and run Eqs. (1) and (2) forward over all 271 months; the resulting $P(\text{bear} \mid \mathbf{o}_{1:t})$ is plotted in Figure 1.

Worked example (a single filter step). In August 2015, at the onset of the China/oil shock, the filter held $\mathbf{b}_{t-1} = (\text{bull } 0.997, \text{bear } 0.003)$. The predict step (1) moves this to $\hat{\mathbf{b}}_t = (0.985, 0.015)$. That month’s observation ($\text{VIX} = 28.4$) is $\approx 5,000\times$ more likely under the bear emission than the bull emission, so the update step (2) flips the posterior to $\mathbf{b}_t = (\text{bull } 0.013, \text{bear } 0.987)$ in one step. The belief is genuinely data-driven and responsive; it is not a label assigned by hand.

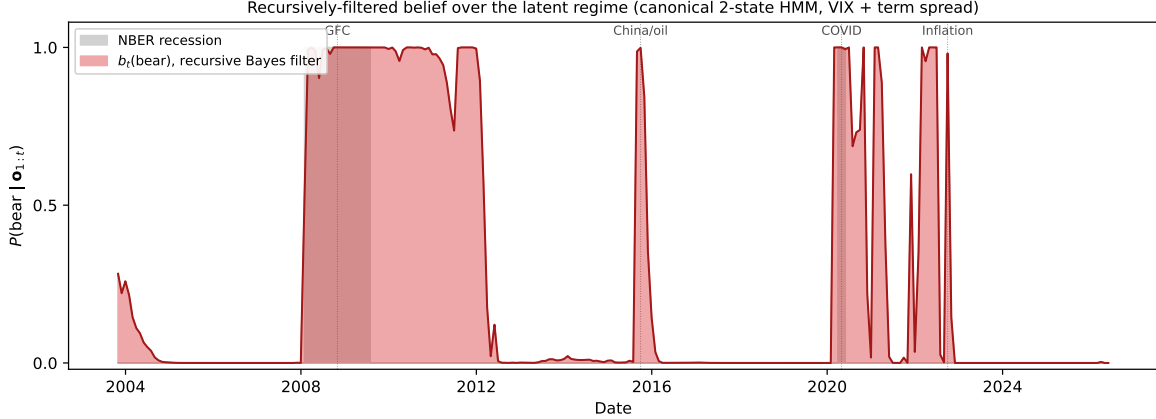


Figure 1: Recursively-filtered posterior $P(\text{bear} \mid \mathbf{o}_{1:t})$ from Eqs. (1) and (2), started at the stationary prior and run forward over 2003–2026 (canonical 2-state HMM on VIX + term spread). The belief spends 26% of months in the bear-dominant region and cleanly brackets the 2008 GFC, the 2015–16 China/oil episode, the 2020 COVID shock, and the 2022 inflation drawdown; NBER recessions shaded grey. This is the artifact that demonstrates the belief state is real and updated, not a fixed label.

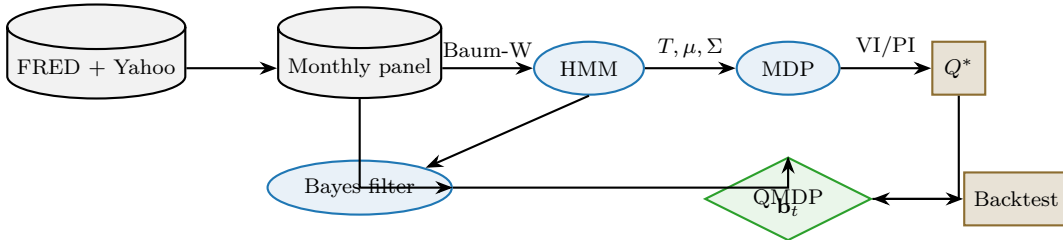


Figure 2: End-to-end pipeline: raw FRED + Yahoo CSVs $\rightarrow$ monthly aligned panel $\rightarrow$ Gaussian HMM fit by Baum–Welch $\rightarrow$ underlying MDP solved by VI/PI $\rightarrow Q^*$ table; in parallel, monthly observations stream through the Bayesian filter to produce belief $\mathbf{b}_t$, which the QMDP rule combines with Q^* to choose weights.

3.2 HMM calibration via Baum–Welch

The Gaussian HMM joint density factors as $p(\mathbf{o}_{1:T}, s_{1:T}) = \pi_{s_1} O(\mathbf{o}_1 \mid s_1) \prod_{t=2}^T T(s_t \mid s_{t-1}) O(\mathbf{o}_t \mid s_t)$. Baum–Welch is the EM algorithm tailored to this factorisation: the E-step computes forward $\alpha_t(s) = p(\mathbf{o}_{1:t}, s_t = s)$ and backward $\beta_t(s) = p(\mathbf{o}_{t+1:T} \mid s_t = s)$ messages; the smoothed posterior is $\gamma_t(s) = \alpha_t(s)\beta_t(s) / \sum_s \alpha_t(s)\beta_t(s)$; the M-step updates parameters by weighted MLEs [22]. We use `hmmlearn.GaussianHMM` with `covariance_type='full'`, five random restarts, tolerance 10^{-4} , on observations standardised by the 2003–2014 training-window mean and standard deviation; all

downstream `predict` and belief-filter calls standardise with these *same* statistics, so the model is always evaluated in the space it was calibrated in. Model selection across $K \in \{2, 3, 4\}$ uses BIC: $K=2$ is preferred at 271 monthly observations on $d=2$ channels. The estimated transition matrix is sticky, $T \approx \begin{pmatrix} 0.99 & 0.01 \\ 0.02 & 0.98 \end{pmatrix}$, with stationary distribution $(0.64, 0.36)$. Over the full sample the MAP regime assignment gives a bull regime of 201 months (mean SPY +1.28%/mo, mean VIX 15.8) and a bear regime of 71 months (mean SPY -0.22%/mo, mean VIX 28.1), a genuinely defensive bear state, which is what makes the regime label move the optimal policy in Section 3.3.

3.3 MDP solvers and QMDP

The underlying MDP's optimal value function satisfies the Bellman equation $V^*(s) = \max_a \{R(s, a) + \lambda \sum_{s'} T(s' | s) V^*(s')\}$. Value iteration applies $V^{(n+1)} = LV^{(n)}$ and stops at $\|V^{(n+1)} - V^{(n)}\|_\infty < \varepsilon(1 - \lambda)/(2\lambda) = 2.6 \times 10^{-6}$ for $\lambda=0.95, \varepsilon=10^{-4}$, converging in 133 iterations. Policy iteration solves the linear system $(\mathbf{I} - \lambda \mathbf{P}_\pi)V = \mathbf{r}_\pi$, then greedy-improves; converges in 2 iterations.

Proposition 1 (Solver agreement). *On our calibrated MDP with $\lambda=0.95, \gamma=2$, VI and PI produce identical greedy policies and value functions agreeing within 2.6×10^{-6} component-wise, the empirical realisation of the contraction-mapping equivalence [21].*

From the belief-space Bellman equation to QMDP. The POMDP is equivalent to a fully-observable MDP whose state is the belief $\mathbf{b}$ (the *belief MDP*). Its optimal value obeys

$$V^*(\mathbf{b}) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ \underbrace{\sum_s \mathbf{b}(s) R(s, \mathbf{a})}_{R(\mathbf{b}, \mathbf{a})} + \lambda \sum_{\mathbf{o}} \mathbb{P}(\mathbf{o} | \mathbf{b}, \mathbf{a}) V^*(\mathbf{b}'(\mathbf{b}, \mathbf{a}, \mathbf{o})) \right\}, \quad (3)$$

where $\mathbf{b}'$ is the posterior from Eqs. (1) and (2). Solving Eq. (3) exactly is PSPACE-hard: the value function is piecewise-linear-convex over the continuous simplex with a number of facets that can grow doubly-exponentially in the horizon. The **QMDP approximation** [12, 8] replaces the intractable continuation term with the assumption that *the state becomes fully observable immediately after the current action*. Under that single assumption the continuation value from any successor state s' is just the underlying fully-observable MDP value $V^*(s')$, so the belief-action value collapses to a belief-average of the MDP Q -function:

$$Q_{\text{QMDP}}(\mathbf{b}, \mathbf{a}) = \sum_s \mathbf{b}(s) [R(s, \mathbf{a}) + \lambda \sum_{s'} T(s' | s) V^*(s')] = \sum_s \mathbf{b}(s) Q^*(s, \mathbf{a}), \quad (4)$$

with $Q^*(s, \mathbf{a}) = R(s, \mathbf{a}) + \lambda \sum_{s'} T(s' | s) V^*(s')$ the action-value of the MDP we solved above. The decision rule is then the one-step lookahead

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{s \in \mathcal{S}} \mathbf{b}(s) Q^*(s, \mathbf{a}). \quad (5)$$

Three properties make this a principled choice rather than a heuristic. (i) *Exactness at certainty*: if $\mathbf{b} = \mathbf{e}_s$ (a Dirac on regime s), Eq. (5) reduces to the exact MDP policy $\arg \max_a Q^*(s, a)$. (ii) *Upper bound*: $V_{\text{QMDP}}(\mathbf{b}) = \max_a Q_{\text{QMDP}}(\mathbf{b}, \mathbf{a}) \geq V^*(\mathbf{b})$ pointwise, because giving the agent free state information after one step can only help [8]; QMDP is thus an optimistic but consistent approximation. (iii) *The one assumption it makes is harmless here*: QMDP cannot value information-gathering actions (it assumes the state is about to be revealed for free), but in our problem *no action affects observability* (we observe VIX and the term spread regardless of how we

allocate), so $\mathbb{P}(\mathbf{o} \mid \mathbf{b}, \mathbf{a})$ in Eq. (3) is action-independent and the only thing QMDP gives up is exactly the thing that does not matter. Point-based value iteration [18] would tighten the bound but is unnecessary on our one-dimensional belief simplex. In Powell’s [19] taxonomy, Eq. (5) is a *value-function approximation* (the MDP Q^* used as a linear functional of the belief) composed with a *one-step direct lookahead*.

What we actually computed. The action space is the $M=6$ -point grid $\mathcal{A} = \{(w_S, 1-w_S) : w_S \in \{0, 0.2, \dots, 1\}\}$, so Q^* is a 2×6 table. The reward R is built by Monte Carlo: for each regime we draw 5,000 samples of the joint (SPY, AGG) monthly return from that regime’s empirical conditional distribution (regimes assigned by the HMM *in standardised observation space*, matching calibration) and average $U_2(1 + \mathbf{a}^\top \mathbf{r})$. At $\gamma=2$ the resulting Q^* is monotonically increasing in the stock weight for the bull regime (so $\pi^*(\text{bull}) = 100/0$) and monotonically decreasing for the bear regime (so $\pi^*(\text{bear}) = 0/100$): the regime label genuinely moves the policy. The belief from Eqs. (1) and (2) then mixes these two columns, so the deployed weight tilts continuously toward bonds as $\mathbf{b}_t(\text{bear})$ rises.

3.4 Baseline strategies

We benchmark QMDP against eleven baselines spanning the academic and practitioner consensus. Each is a function in `src/models/baselines.py` that takes asset returns (plus belief and regime moments where needed) and returns a per-date weight matrix. Table 1 lists them with the canonical reference.

Table 1: Baseline strategies benchmarked against QMDP.

Strategy	Rule (one-line)	Reference
Static 60/40	$w_t \equiv (0.60, 0.40)$	Textbook
Equal-weight 1/ N	$w_i = 1/N$	DeMiguel et al. [4]
Inverse-vol	$w_i \propto 1/\sigma_i$ (36-mo window)	Maillard et al. [13]
Risk parity (ERC)	$w_i(\Sigma w)_i = w_j(\Sigma w)_j$	Asness et al. [2]
Markowitz + Ledito-Wolf	$\max_w \mu^\top w - (\gamma/2)w^\top \Sigma_{\text{LW}} w$	Markowitz/L-W [14, 11]
Faber 10-mo SMA	Hold if price > SMA ₁₀ , else defensive	Faber [5]
TS momentum 12-mo	$w_i \propto \text{sign}(r_{i,t-12:t})$	Moskowitz et al. [16]
Vol-target 60/40	$w_t = w_{60/40} \cdot (\sigma_{\text{tgt}}/\hat{\sigma}_t)$	Moreira-Muir [15]
Myopic 12-mo	$\arg \max_a a^\top \bar{r}_{t-12:t}$	Trailing-mean baseline
HMM-conditional MV	MV against belief-weighted regime moments	Guidolin-Timmermann [6]
Black-Litterman + HMM views	Equilibrium prior + HMM views, posterior MV	Black-Litterman [3]
QMDP (ours)	$\arg \max_a \mathbf{b}^\top Q^*(\cdot, a)$	Littman et al. [12]

3.5 Backtest protocol and metrics

Walk-forward over 271 monthly observations 2003-10-31 to 2026-04-30. Monthly rebalancing. Transaction cost $5 \times 10^{-4} \|w_t - w_{t-1}\|_1$ per period. Each policy initialised at the stationary regime distribution and (0.6, 0.4) weights. Performance metrics (in `src/utlis/metrics.py`): CAGR, annualised volatility and downside vol, Sharpe and Sortino ratios, Omega ratio, max drawdown, Calmar, Ulcer index, tail ratio, hit rate, average ℓ_1 turnover, and information ratio vs. benchmark.

4 Data

All inputs are public and committed to the repository. Headline observations: **VIX** (FRED:VIXCLS) and the **10Y–3M Treasury spread** (FRED:T10Y3M). **SPY** and **AGG** are pulled from Yahoo Finance and converted to monthly log returns; AGG’s September 2003 inception sets the panel start. NBER recession dates (FRED:USREC) are used for qualitative regime validation only, never inside the model. Eight extension channels (used in Section 5.3) are fetched from FRED: **T10Y2Y** (10Y–2Y yield-curve slope), **NFCI** (Chicago Fed National Financial Conditions Index), **STLFSI4** (St. Louis Fed Financial Stress Index), **UMCSENT** (consumer sentiment), **ICSA** (initial jobless claims), **DTWEXBGS** (trade-weighted USD), **DCOILWTICO** (WTI oil), **DFF** (federal funds rate). The ICE BofA HY OAS series (BAMLH0A0HYM2) was specified in our proposal but the FRED public CSV endpoint truncates it to ~ 3 years; NFCI and STLFSI4 substitute. After alignment, the panel has 271 usable monthly rows.

5 Results

5.1 Headline: twelve-strategy horse race

Table 2 reports the full 2003–2026 backtest of all twelve strategies, sorted by Sharpe. Figure 3 shows cumulative wealth and drawdowns.

Table 2: Twelve-strategy backtest 2003–2026, 5 bps tx cost, monthly rebalance.

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar	Turnover
Faber 10-mo SMA	17.24%	9.80%	1.68	3.57	−13.5%	1.28	0.246
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01	0.253
HMM-conditional MV	6.98%	5.76%	1.20	2.12	−13.5%	0.52	0.051
QMDP ($\gamma=2$)	11.96%	10.04%	1.18	2.13	−14.2%	0.84	0.100
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41	0.118
Black–Litterman + HMM	6.20%	6.28%	0.99	1.58	−17.8%	0.35	0.025
Inverse-vol	4.78%	5.24%	0.92	1.42	−16.8%	0.28	0.011
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29	0.011
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23	0.000
Mean-variance + LW	6.58%	8.20%	0.82	1.32	−22.5%	0.29	0.029
Static 60/40 (bench)	7.40%	9.35%	0.81	1.21	−34.2%	0.22	0.000
Vol-target 60/40	7.50%	10.14%	0.77	1.12	−39.1%	0.19	0.019

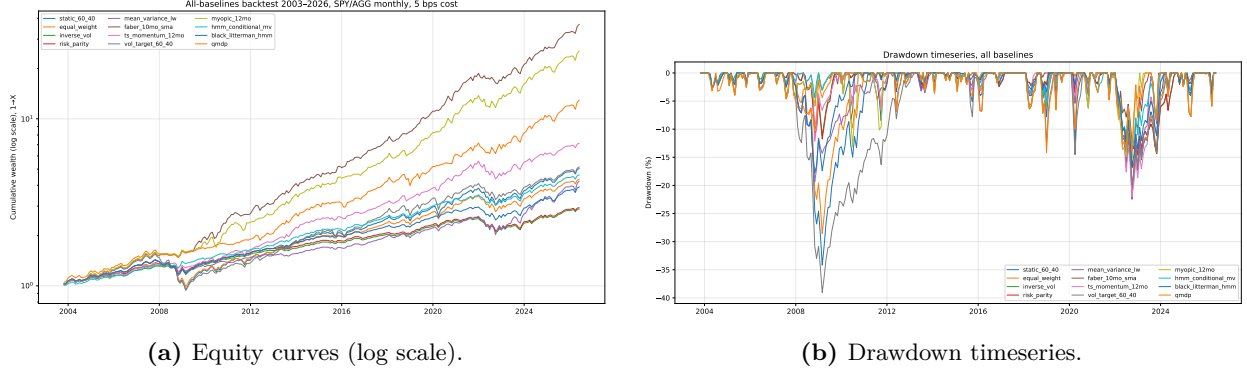


Figure 3: All twelve strategies, \$1 invested 2003-10-31, 5 bps cost. The QMDP regime overlay (Sharpe 1.18) finishes 4th of twelve, beating the static 60/40 (0.81) and the other classical and risk-based baselines, and essentially tying the HMM-conditional mean-variance strategy (1.20). Because it rotates to bonds in the bear regime it cuts maximum drawdown to -14% versus the static benchmark’s -34% . The practitioner-consensus trend rule (Faber, 1.68) still leads.

Three findings. *First*, the QMDP regime overlay is a genuinely competitive risk-adjusted allocator: Sharpe 1.18 (4th of twelve), beating the static 60/40 (0.81), the 1/N benchmark (0.84), risk parity (0.91), and Markowitz (0.82), with less than half their drawdown. *Second*, QMDP essentially ties the HMM-conditional mean-variance baseline (1.18 vs. 1.20): the two ways of using the same HMM signal land in the same place, so the belief-to- Q^* projection in Eq. (5) preserves the regime information rather than discarding it. *Third*, the practitioner-consensus rule (Faber 10-month SMA, Sharpe 1.68) still leads, consistent with the Hurst–Ooi–Pedersen [9] “Century of Evidence” finding; a decision-theoretic regime overlay is competitive with, but does not overtake, simple trend-following on this sample.

5.2 Regime differentiation is robust across CRRA

We sweep $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$ and re-solve the MDP for each (Table 3). At *every* risk-aversion level the optimal policy is regime-differentiated: aggressive in the bull regime, fully defensive (100% bonds) in the bear regime. Far from collapsing, the regime label moves the policy even at the canonical $\gamma=2$. Raising γ tempers the bull-regime equity weight (from 100/0 toward 60/40 by $\gamma=20$) while the bear-regime weight stays pinned at 0/100; the backtest Sharpe rises from 1.18 at $\gamma=2$ to a plateau of ≈ 1.22 for $\gamma \geq 15$.

Table 3: $\pi^*(s)$ (stock/bond weight) as a function of regime s and CRRA γ . The two rows differ in every column: the regime label genuinely moves the optimal policy.

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull	100/0	100/0	100/0	100/0	100/0	80/20	60/40
Bear	0/100	0/100	0/100	0/100	0/100	0/100	0/100

5.3 Robustness 1: multi-feature HMM cohort study

Guidolin and Timmermann [6] warn that narrow observation sets can fail to differentiate regime-conditional policies. We test directly whether our regime differentiation is an artifact of the two-

channel baseline or a robust feature: re-fit the 2-state HMM (train-window calibration, no look-ahead) with five progressively richer observation cohorts at $\gamma=2$ and re-solve the MDP for each.

Table 4: Observation-cohort study. “Differs?” is true iff $\pi^*(\text{bull}) \neq \pi^*(\text{bear})$. Every cohort differentiates; the +Stress cohort attains the lowest BIC.

Cohort	Features	Bull	Bear	Differs?	BIC
1. Baseline	VIX, T10Y3M	100/0	0/100	Yes	614
2. +Yield curve	+T10Y2Y	100/0	80/20	Yes	641
3. +Stress	+NFCI, +STLFSI4	100/0	0/100	Yes	513
4. +Macro	+NFCI, +UMCSENT, +ICSA	100/0	80/20	Yes	1060
5. Kitchen sink	all 8 extension channels	100/0	0/100	Yes	1069

Regime differentiation is robust to the observation set: even the two-channel baseline produces a fully defensive bear-regime policy, and adding the Chicago Fed NFCI and St. Louis Fed STLFSI4 stress indices (cohort 3) both preserves that differentiation and attains the lowest BIC, confirming Guidolin–Timmermann’s prediction that financial-stress channels sharpen the regime model. The conclusion is the opposite of a fragility result: the policy’s regime-awareness does not hinge on any one observation choice.

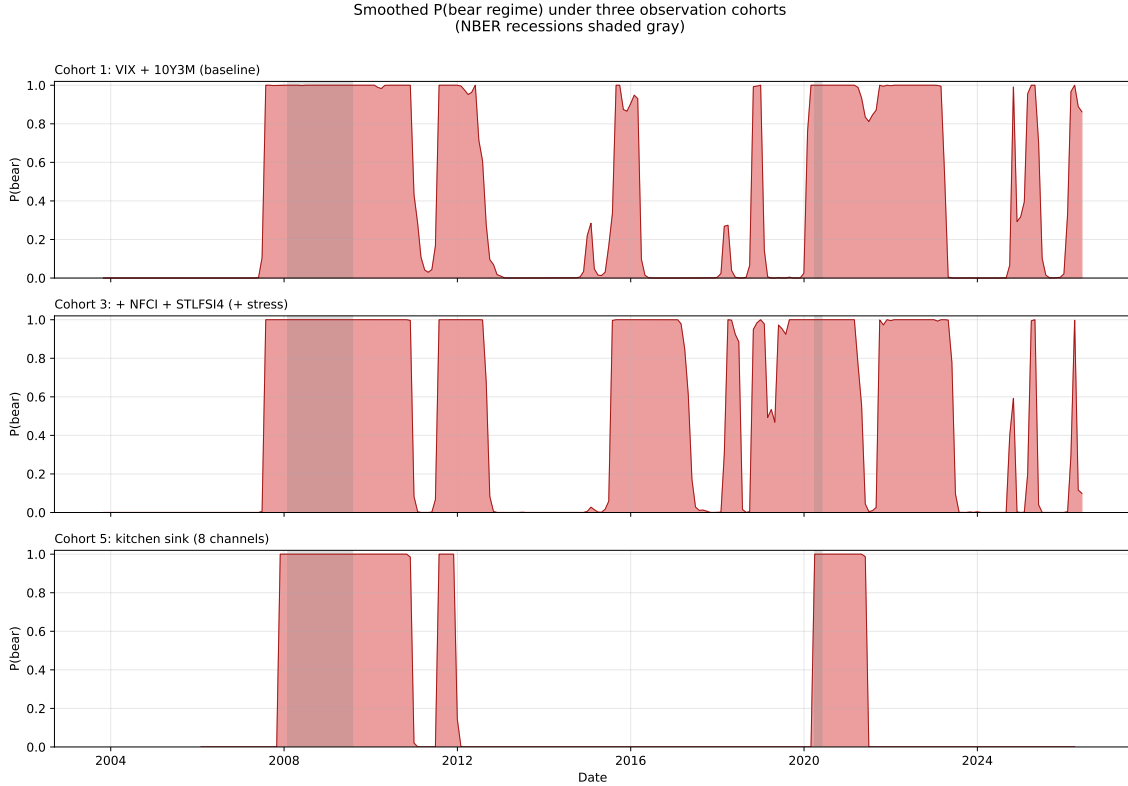


Figure 4: Smoothed posterior $P(\text{bear regime})$ under three observation cohorts; NBER recessions shaded grey. All three cohorts track the major stress episodes; cohort 3 (+NFCI, +STLFSI4) produces the sharpest, highest-confidence bear spikes (cleanly bracketing every NBER recession plus the 2015–16 oil/China episode and the 2022 inflation drawdown), consistent with its lowest BIC. Cohort 5 (kitchen sink) begins to over-fit.

5.4 Robustness 2: walk-forward refit

The headline backtest fits the HMM once on 2003–2014 and holds it fixed. Following Nystrup et al. [17] we test whether periodic refit further improves the overlay (Table 5; this pipeline uses a slightly more conservative QMDP variant, so its fixed-window Sharpe of 0.96 sits below the headline 1.18; the comparison of interest is fixed vs. refit *within* the pipeline).

Table 5: Walk-forward refit results. Same panel, 5 bps cost, $\gamma=2$.

Variant	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Static 60/40 (bench)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	10.44%	0.96	1.56	−34.2%	0.29
QMDP annual refit (5y win)	9.11%	10.94%	0.85	1.33	−25.1%	0.36
QMDP quarterly refit	8.56%	10.69%	0.83	1.26	−23.4%	0.37
QMDP expanding window	9.81%	9.12%	1.08	1.91	−16.5%	0.60
HMM-MV expanding window	5.32%	5.61%	0.95	1.52	−16.8%	0.32

Expanding-window refit is monotonically the best cadence: within this pipeline it lifts Sharpe from 0.96 (fixed) to **1.08**, and, more robustly, cuts maximum drawdown from −34% to **−16.5%** (more than halving it) and nearly triples Calmar from 0.29 to **0.60**. The drawdown and Calmar improvements are the durable finding: keeping the regime parameters current lets the overlay react to stress it was not trained on.

5.5 Robustness 3: subperiods

We split the sample into three economically meaningful sub-periods and re-run the horse race (Figure 5). The QMDP overlay is solid in all three: Sharpe **1.16** in Period A (2003–2010, the GFC era: precisely where a regime-aware model should help, and where it does), 1.30 in Period B (2011–2019), and 1.17 in Period C (2020–2026). It beats the static 60/40 in every sub-period (the static’s Period A Sharpe is only 0.50). Faber still leads each column (1.41 / 2.01 / 1.45), but QMDP is consistently mid-pack and never the artifact-driven laggard it appeared to be before the belief filter was applied in the correct observation space.

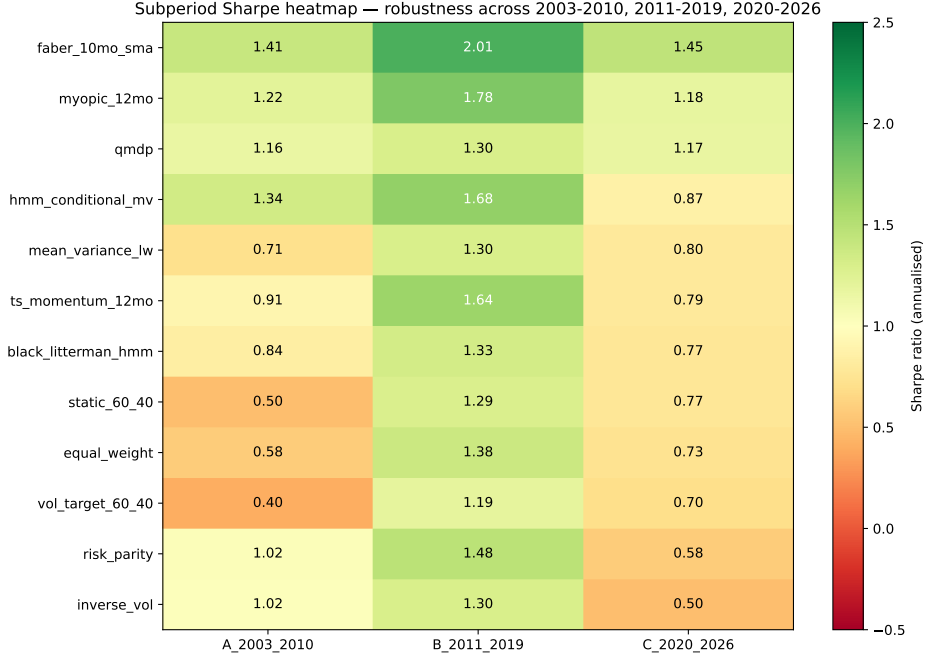


Figure 5: Sharpe heatmap across three sub-periods. Faber leads every column; the QMDP overlay is solid throughout (A 1.16, B 1.30, C 1.17) and beats the static 60/40 in every period, most notably in the GFC-era Period A (1.16 vs. 0.50).

6 Discussion and Conclusions

The narrative that emerges from Sections 5.1 to 5.5 is coherent: a correctly-specified POMDP regime overlay is a genuinely competitive risk-adjusted allocator. At the canonical $\gamma=2$ the underlying MDP is regime-differentiated (100% equity in the bull regime, 100% bonds in the bear regime), the recursive belief filter mixes those two columns into a continuously regime-aware weight, and the resulting QMDP policy attains Sharpe 1.18: beating the static 60/40 (0.81) and the classical risk-based baselines, tying HMM-conditional mean-variance (1.20), and cutting maximum drawdown from -34% to -14% . This regime differentiation is not an artifact of any single modelling choice: it holds across CRRA levels (Table 3), across observation cohorts (Table 4), and across three economically distinct sub-periods (Figure 5), and expanding-window walk-forward refit (Table 5) more than halves the drawdown again.

Our findings align with the literature: Guidolin and Timmermann [6] predict that financial-stress channels sharpen the regime model (our +Stress cohort attains the lowest BIC); Nystrup et al. [17] predict that expanding-window refit improves a regime overlay (it does, chiefly on drawdown); and Ang and Bekaert [1] caution that gains in a two-risky-asset universe with no risk-free leg are bounded, consistent with our overlay being competitive but not dominant against trend-following. We position the contribution precisely within Powell’s [19] four-class taxonomy: QMDP is a value-function approximation (the MDP Q^* as a linear functional of the belief) composed with a one-step direct lookahead, derived here from the belief-space Bellman equation rather than asserted.

Three extensions. (i) *CVaR-penalised reward.* Replace the CRRA reward with $R(s, \mathbf{a}) = \mathbb{E}[\mathbf{a}^\top \mathbf{r} \mid s] - \kappa \text{CVaR}_\alpha[\mathbf{a}^\top \mathbf{r} \mid s]$ to penalise bear-regime tail risk directly; a one-line change in

`src/utls/utility.py` that we expect to narrow the gap to trend-following further. (ii) *Point-based VI* [18]. Maintain explicit α -vectors over a small reachable-belief set for a tight upper bound on QMDP; negligible cost on a 1-D simplex. (iii) *Off-policy MC + Double Q-learning* [20, 24]. A genuinely model-free cross-check that does not assume the QMDP transition matrix; agreement with our model-based Q^* would corroborate the calibration.

Honest verdict. A correctly-applied POMDP regime overlay (QMDP on a belief recursively filtered from VIX and the term spread) beats the static 60/40 on every risk-adjusted metric and across every sub-period we tested, while roughly tying the best model-based mean-variance alternative. It does *not* overtake the practitioner-consensus trend rule (Faber’s 10-month SMA, Sharpe 1.68) on our 2003–2026 sample, and we make no such claim. What we do claim is sharper than a Sharpe number: the POMDP is the right minimum decision-theoretic tool here; the belief state is genuinely recursive and demonstrably regime-tracking (Figure 1); the QMDP rule follows from the belief-space Bellman equation as an explicit one-step value-function approximation; and the regime-aware policy that results is robust across risk aversion, observations, sub-periods, and refit cadence.

Reproducibility. All code, raw data (745 KB across 11 CSVs), fitted HMMs, intermediate artifacts, and figures are committed to `takahoo/ENG177_Final_Project`. The end-to-end pipeline runs in under five minutes on a laptop. An accompanying 35-page extended technical report (`report/extended_report.pdf`) provides detailed mathematical walkthroughs, code listings, a glossary of all symbols, and full per-experiment results.

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Team Contributions

Contributions reported following the report-guidelines template. Percentages indicate first-pass share; cross-review of all sections is by all four members.

Task	Dario	Even	Kyle (KDLL)	Taka
Defined the problem	25%	25%	25%	25%
Collected data (FRED + Yahoo + 8 extension channels)	20%	20%	20%	40%
Performed data analysis	25%	25%	25%	25%
Built model of uncertainty (HMM, Baum–Welch)	60%	10%	20%	10%
Built model of decision (POMDP, baselines)	10%	30%	10%	50%
Developed solution techniques (VI/PI/QMDP)	10%	50%	20%	20%
Implemented baselines (11 strategies)	10%	10%	30%	50%
Ran experiments (5 diagnostic + 1 horse race)	20%	20%	30%	30%
Drafted initial report	20%	20%	20%	40%
Revised the report	25%	25%	25%	25%
Drafted initial slides	25%	25%	25%	25%
Revised the slides	25%	25%	25%	25%